Lake Powell and Lake Mead Historical and Current Reservoir Data

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Overview

This is an R Markdown document. The document generates plots for Lake Powell and Lake Mead:

1. Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

Data wrangling strategy

1. Import the reservoir elevation-volume curves.
2. Import Daily, Monthly, and Annual Lake Powell and Lake Mead data from the Reclamation's HydroData Web portal https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html
3. Filter the reservoir data for the appropriate variables/fields.

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

Requested Citation

David E. Rosenberg (2026), "Lake Powell and Lake Mead Historical and Current Reservoir Data." Utah State University. Logan, Utah. https://github.com/dzeke/ColoradoRiverCollaborate/tree/main/HistoricalCurrentReservoirData.

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Figure 1: Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

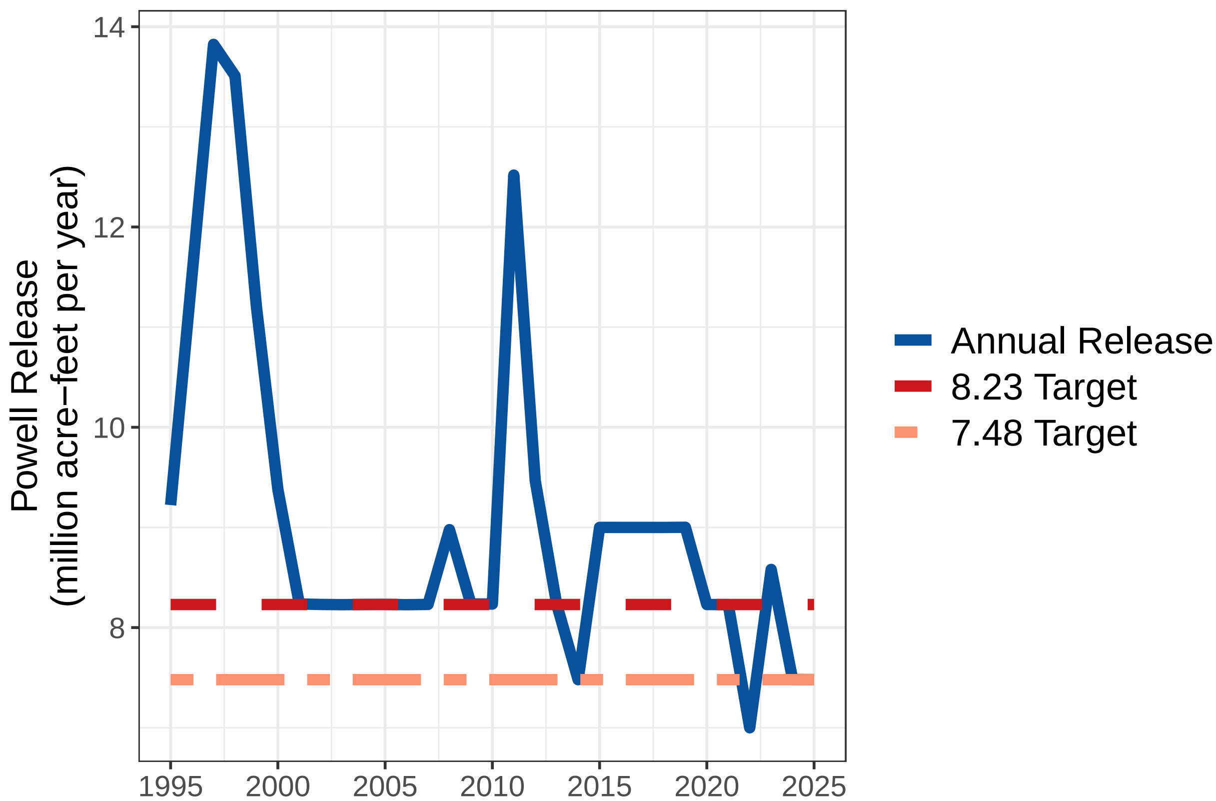

Figure 1: Ten-Year Powell Release Compared to 10-year Targets

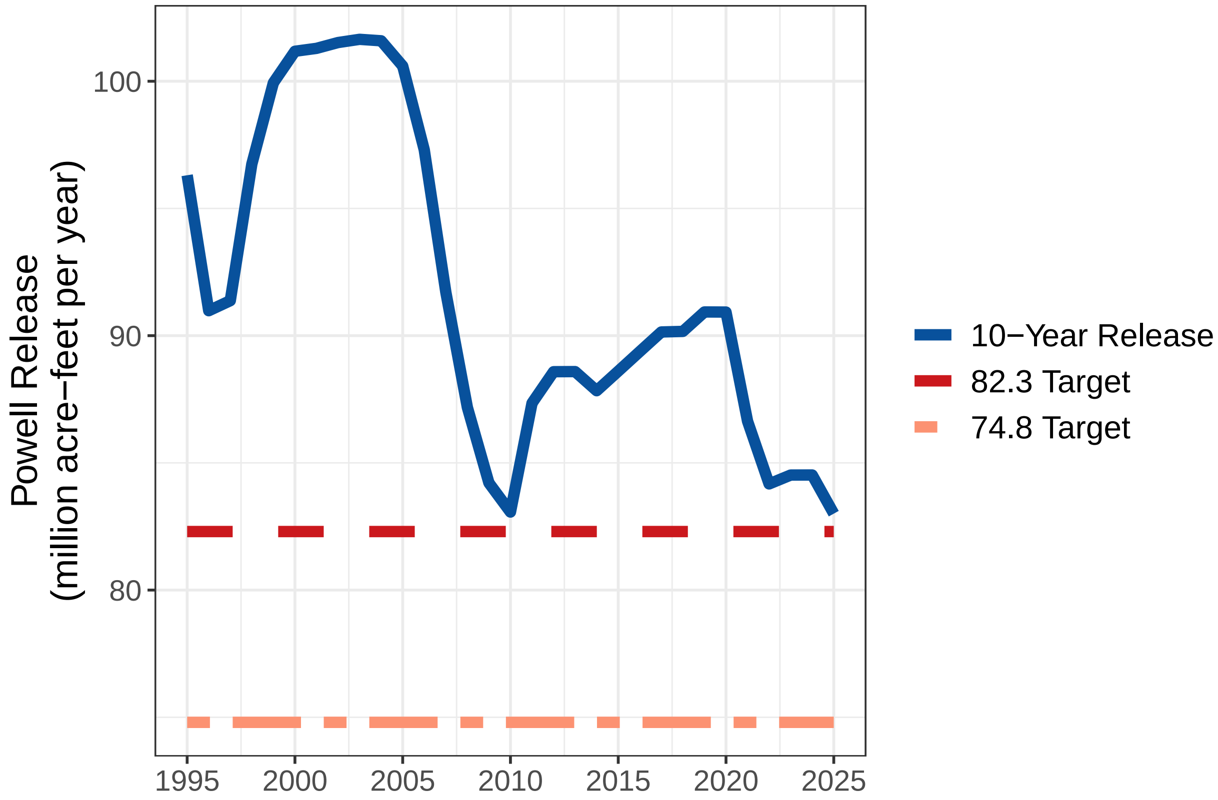

Table 1: Current Storage and Elevations

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[1] "Reservoir Data on: July 20 , 2026"
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ResName	Storage	Pool Elevation
Lake Powell	5.45	3523.29
Lake Mead	7.10	1041.73

Figure 3: Lake Powell, Lake Mead, and Combined Storage

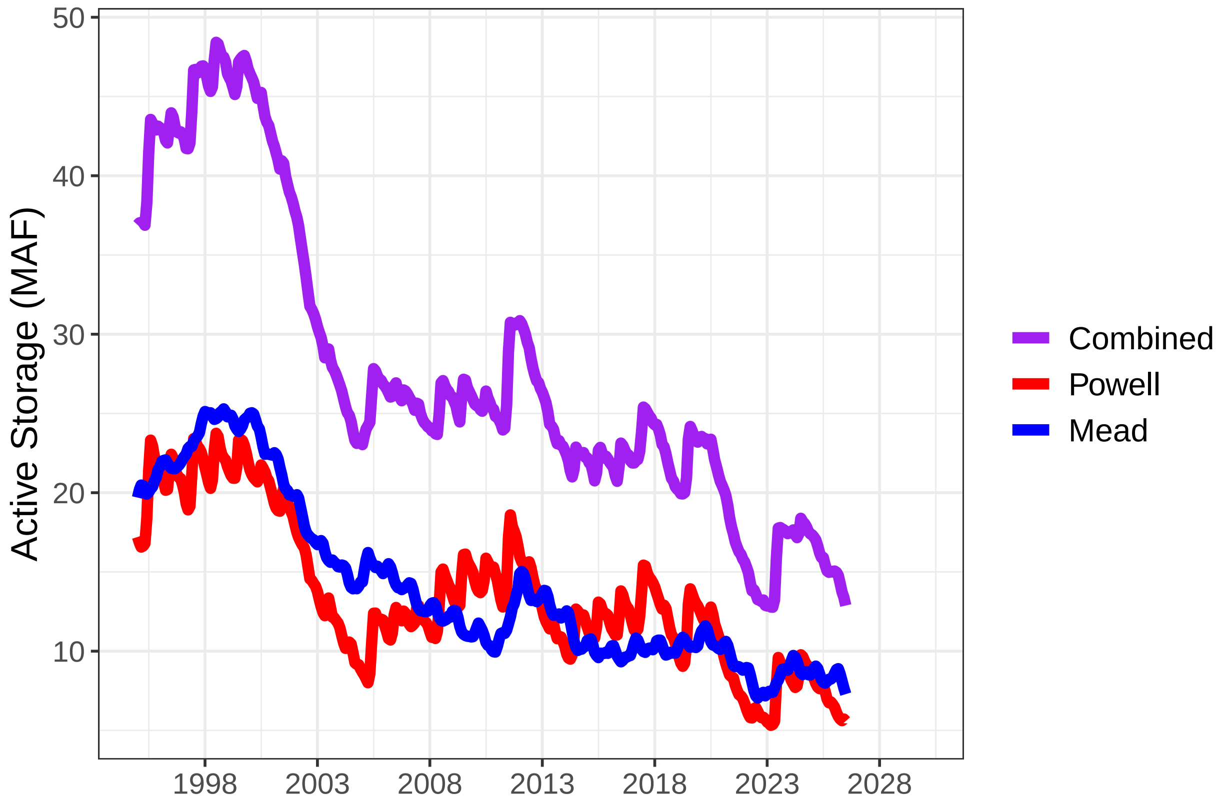


Figure 4: Lake Powell Storage vs Lake Mead Storage

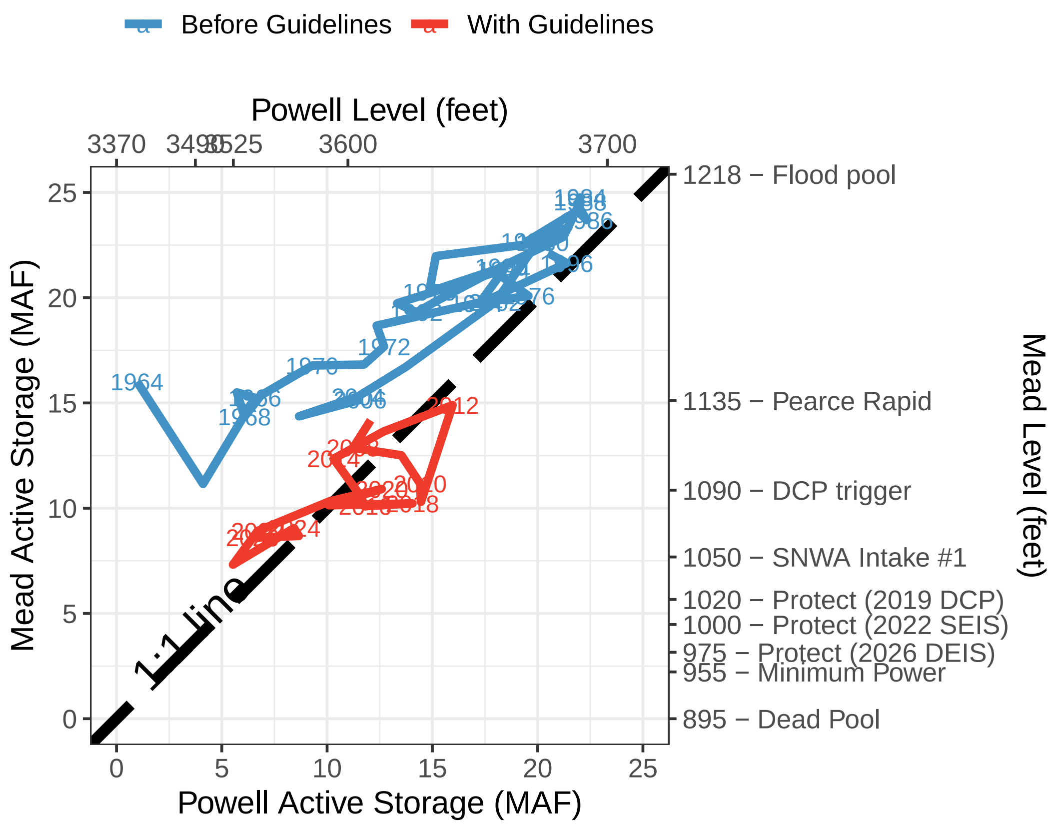
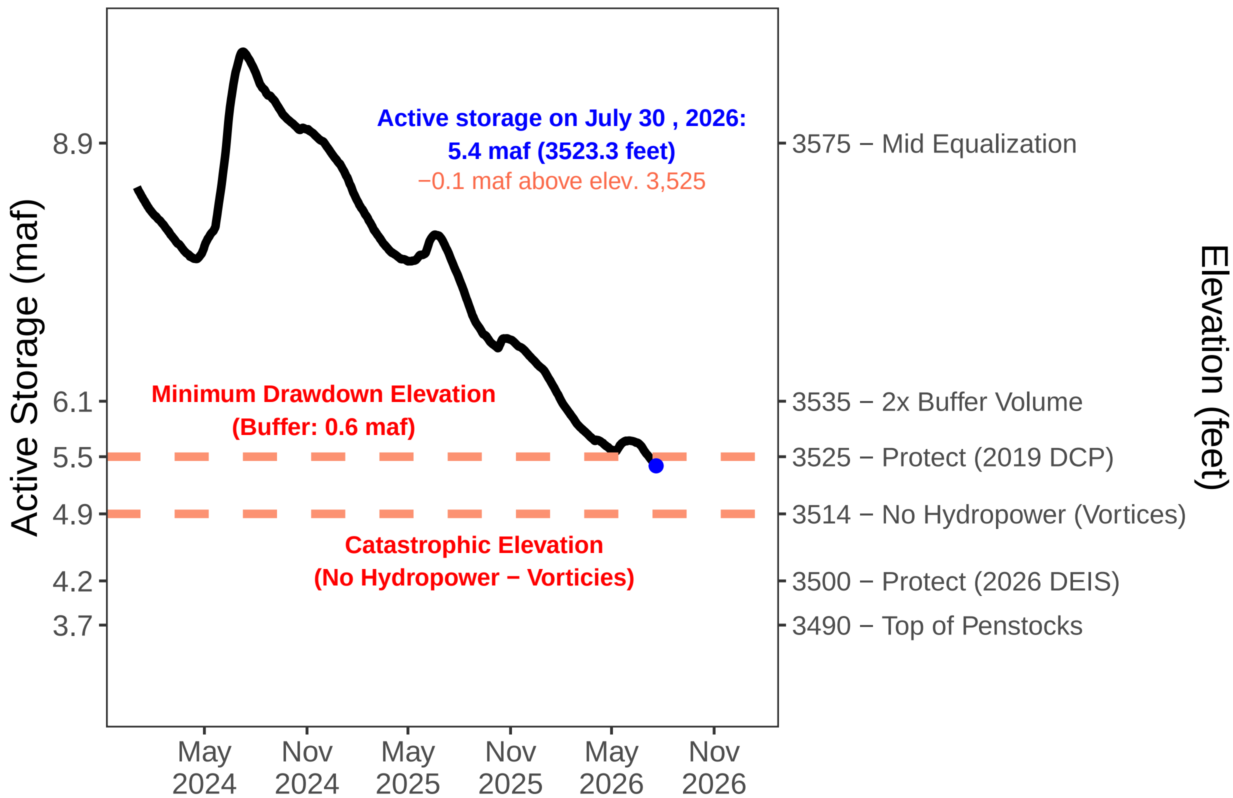


Figure 5: Lake Powell Storage and Elevation over Time.

Figure 5B: Lake Powell Minimum Drawdown and Catastrophic Elevations - Daily Data



Recent Lake Powell Decline 5000 acre-feet per day

Recent Lake Powell Decline 33000 acre-feet per week

Table 2. Time Until Powell Elevations

Label	ActiveStorageMAF	DaysUntilVolume	WeeksUntilVolume
3370 - River Outlets	0.00	1151	164
3490 - Top of Penstocks	3.74	360	51
3500 - Protect (2026 DEIS)	4.22	260	37
3514 - No Hydropower (Vortices)	4.93	109	16
3525 - Protect (2019 DCP)	5.54	-21	-3
3535 - 2x Buffer Volume	6.14	-146	-21
3575 - Mid Equalization	8.90	-730	-104
3600 - Cold Release Temp.	10.99	-1171	-167
3700 - Capacity	23.31	-3775	-539

Figure 6: Lake Mead Storage and Elevation over Time.

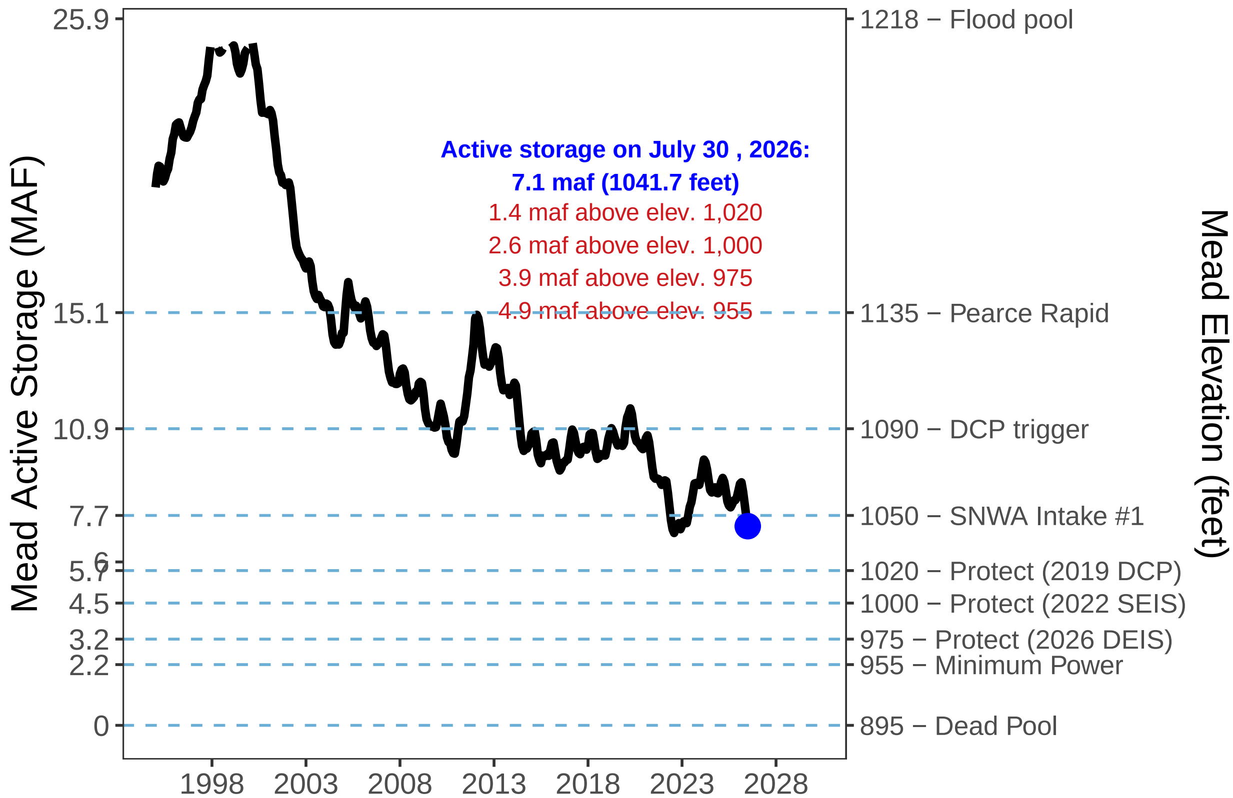


Figure 6B: Lake Powell Daily Inflow and Release Water Year to Date

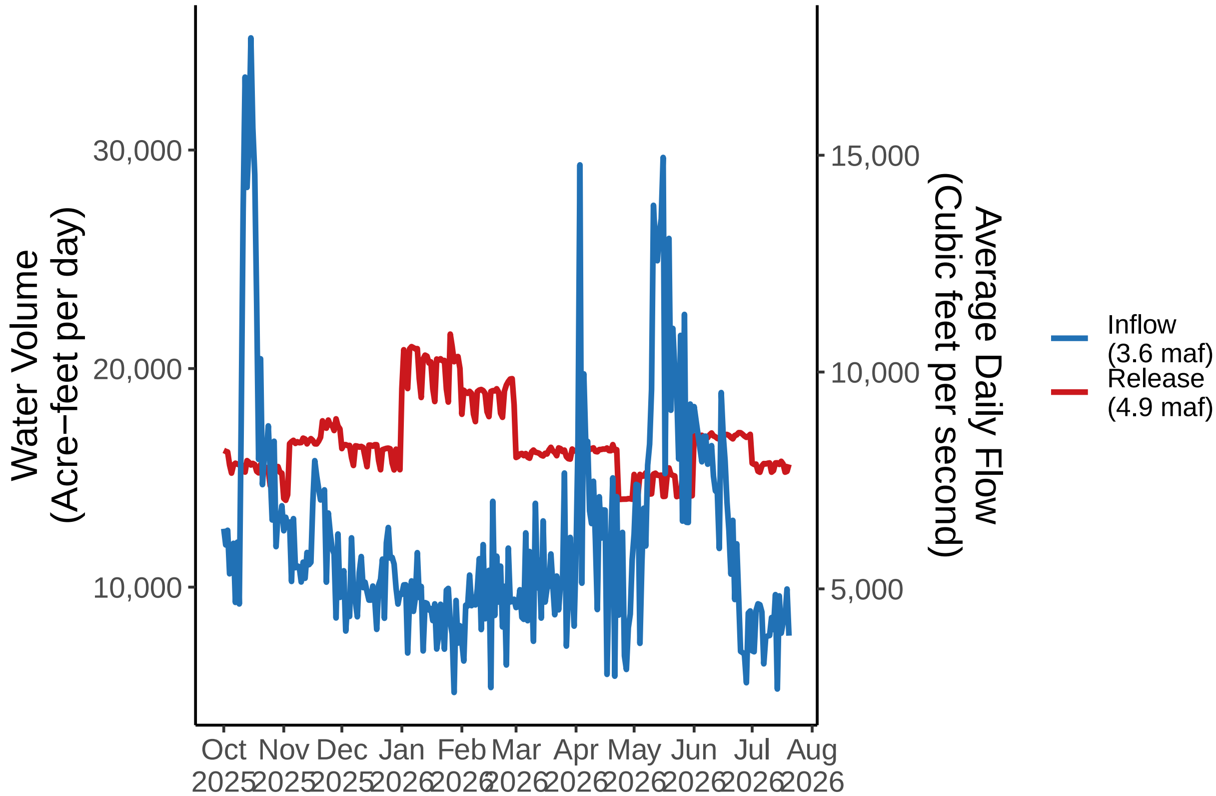


Figure 7: Frequency of Annual Lake Powell Release Volumes.

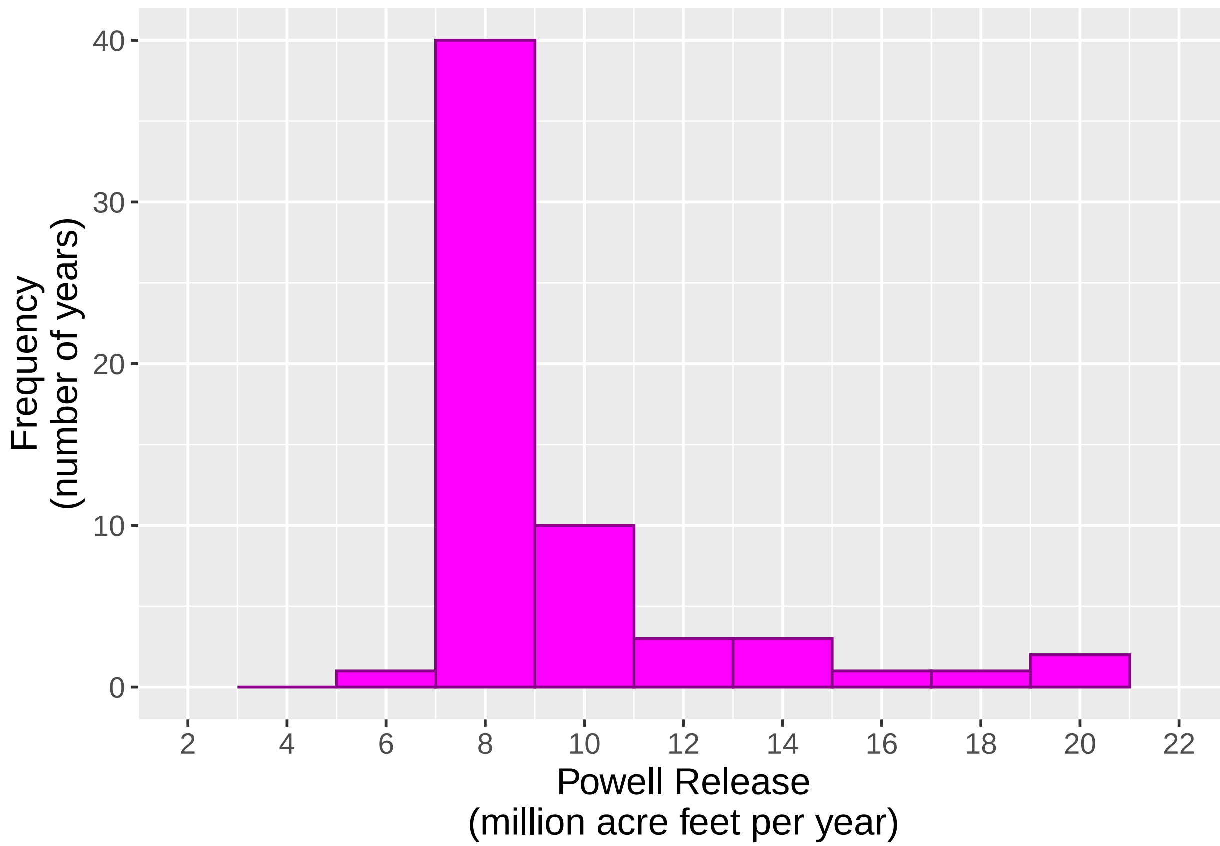


Figure 8: Frequency of Annual Lake Powell Inflow Volumes.

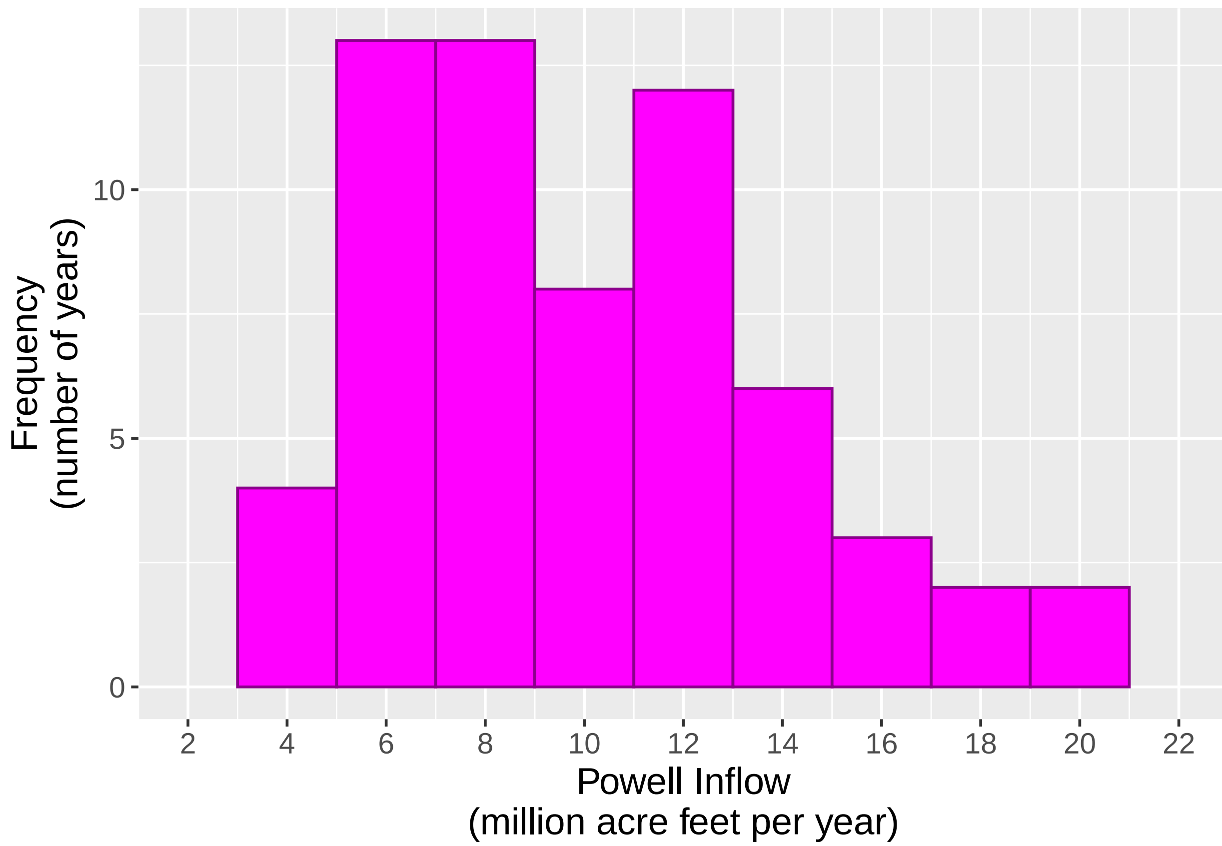


Figure 9: Lake Powell Inflows and Releases.

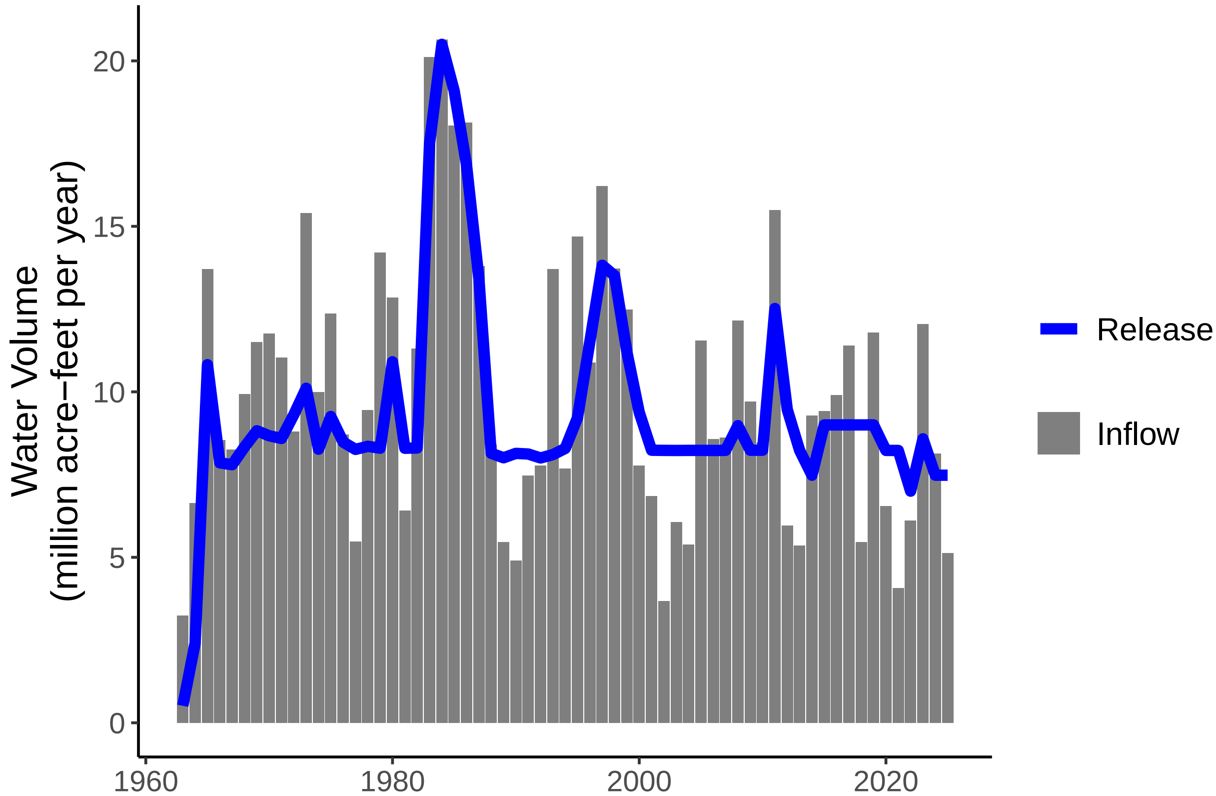


Figure 10: Annual Lake Powell Evaporation.